

Question				Marking details						Marks Available																													
										AO1		AO2		AO3		Total		Maths		Prac																			
3	(a)	(i)		Purple White and Aa aa (1) Aa Aa aa aa (1) correct completion of punnett square							2			2																									
		(ii)		Null hypothesis: There is no <u>significant</u> difference between the observed and expected numbers/ratios (1) <table><tr><td>Phenotype</td><td>O</td><td>E (1)</td><td>O-E</td><td>(O-E)²</td><td>$\frac{(O-E)^2}{E}$ (1)</td></tr><tr><td>white</td><td>32</td><td>25</td><td>7</td><td>49</td><td>1.96</td></tr><tr><td>purple</td><td>18</td><td>25</td><td>-7</td><td>49</td><td>1.96</td></tr></table> Chi ² = 3.92 (1)						Phenotype	O	E (1)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$ (1)	white	32	25	7	49	1.96	purple	18	25	-7	49	1.96		4			4			4			4	
Phenotype	O	E (1)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$ (1)																																		
white	32	25	7	49	1.96																																		
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		(iii)		Critical value is 3.84 (1) because chi ² value {is greater than/to the right of critical value (at 0.05)} (1) reject null hypothesis (1) deviation is not due to chance/there is a significant difference between O and E(1) ECF from (ii)										4			4																						
	(b)	(i)		<table><tr><td></td><td>AB</td><td>Ab</td><td>aB</td><td>ab</td></tr><tr><td>aB</td><td>AaBB</td><td>AaBb</td><td>aaBB</td><td>aaBb</td></tr><tr><td>ab</td><td>AaBb</td><td>Aabb</td><td>aaBb</td><td>aabb</td></tr></table> Gametes (1) Correct completion (1) [from their gametes] Phenotype ratio 3 purple : 5 white (1) [from table]							AB	Ab	aB	ab	aB	AaBB	AaBb	aaBB	aaBb	ab	AaBb	Aabb	aaBb	aabb		3			3										
	AB	Ab	aB	ab																																			
aB	AaBB	AaBb	aaBB	aaBb																																			
ab	AaBb	Aabb	aaBb	aabb																																			
		(ii)		Hypothesis 2 and The chi squared value is {to the left/less than} the critical value/no <u>significant</u> difference between <u>O</u> and <u>E</u> /probability is between {0.8-0.9/above 0.8/above 5%}/due to chance alone										1			1																						

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			Hypothesis 1 white x white could only produce white offspring/hypothesis 2 white x white could give purple/or description with genotypes			1	1		1
				Question 3 total	0	9	6	15	9	9